Year 9 Mathematics

Unit 4: Constructions

4.2 Constructions 1

Constructions 1

STARTER · 5 MIN

Date: _____ **Title: Constructions 1**

LEARNING OBJECTIVES

- Construct the perpendicular bisector of a line
- Construct a perpendicular at or from a point
- Construct a right-angled triangle and spot errors

STARTER *Answer in your book. No calculators.*

1. How many degrees in a right angle?
2. A line is 9 cm. How far is the midpoint from each end?
3. What does perpendicular mean?
4. What is half of 7.4 cm?

Starter — Answers

STARTER

Tick or correct in green pen

1. **90°**
2. **4.5 cm**
3. **at right angles (90°)**
4. **3.7 cm**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can construct the perpendicular bisector of a line	☐
SILVER	I can construct a perpendicular at or from a point	☐
GOLD	I can construct a right-angled triangle and spot errors	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Constructions 1

INTRODUCTION · 3 MIN

Keep the compasses at the same width for each pair of arcs.

Perpendicular bisector

Cuts a line in half at 90°.

Arcs above and below from both ends.

At a point P

Mark Q and R either side of P first.

Then bisect QR.

From a point D

Arc from D cuts the line at Q and R.

Then arcs below from Q and R.

Right-angled triangle

Construct 90°, then arc the hypotenuse.

Leave all arcs showing.

Examples 1 and 2 — Perpendicular Bisectors

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint compasses more than half the line
same width from both ends

Example 1

AB is 6 cm. Construct the perpendicular bisector of AB.



Example 2

PQ is 5 cm. Construct the midpoint M of PQ.
Measure PM.



Examples 1 and 2 — Corrections

CORRECTIONS

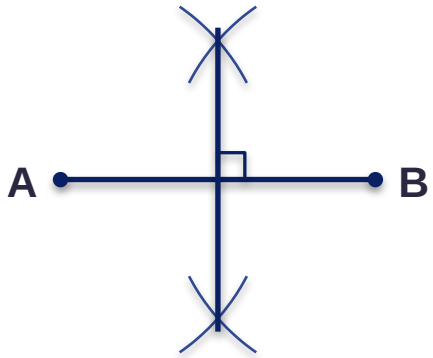
BRONZE

Check your copy and correct in green pen

Example 1

Open more than half of AB

Same arcs from A and from B

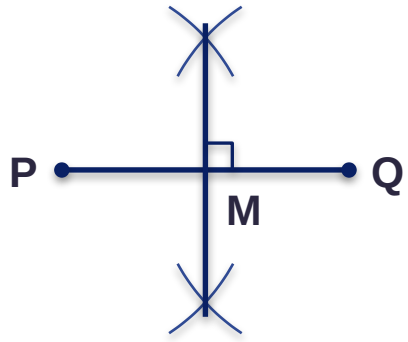


Tip: the line crosses AB at 90°.

Example 2

Bisect PQ: M is where it crosses

PM = 2.5 cm



Tip: M is the midpoint, so PM = MQ.

Task 1

PRACTICE · 9 MIN

BRONZE

Draw each line. Construct its perpendicular bisector. Measure to check.

1. AB = 6 cm
2. AB = 10 cm
3. AB = 7 cm
4. AB = 9.4 cm
5. AB = 8 cm: mark the midpoint M

★ AB = 11 cm: mark the midpoint M

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **3 cm each side, 90°**
2. **5 cm each side, 90°**
3. **3.5 cm each side, 90°**
4. **4.7 cm each side, 90°**
5. **AM = 4 cm**

★ **AM = 5.5 cm**

Examples 3 and 4 — Perpendiculars at and from a Point

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint point on P: mark Q and R first then bisect QR

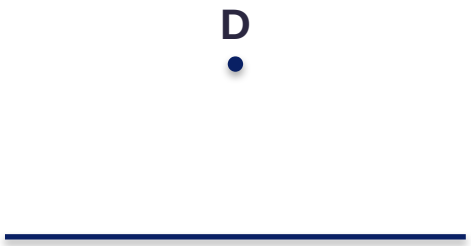
Example 3

Construct the perpendicular to AB at the point P.



Example 4

Construct the perpendicular from D to the line.



Examples 3 and 4 — Corrections

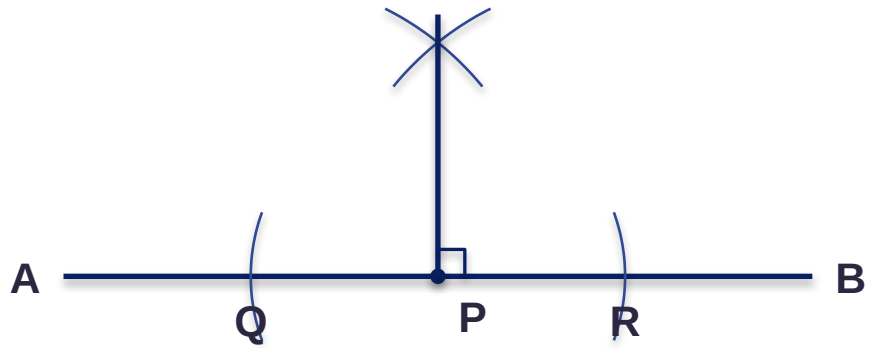
CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

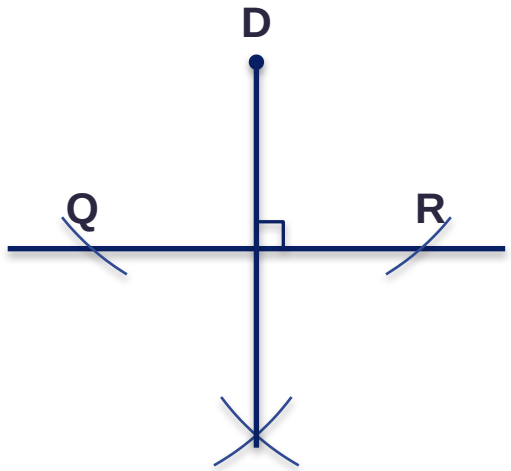
Mark Q and R, then bisect QR



Tip: the point of the compasses goes on P.

Example 4

Arc from D, then arcs below



Tip: extend the line if the arc misses it.

Task 2

PRACTICE · 9 MIN

SILVER

Draw the diagram, then construct the perpendicular. Measure to check.

1. Line AB = 10 cm, P is 4 cm from A. Perpendicular at P.

2. Line AB = 8 cm, P is 5 cm from A. Perpendicular at P.

3. Line AB = 9 cm, P is the midpoint. Perpendicular at P.

4. D is 3 cm above a line. Perpendicular from D to the line.

5. D is 4 cm above a line. Perpendicular from D to the line.

★ Triangle: base 7, sides 5, 6 cm. Perpendicular from the top.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. angle at P = 90°
2. angle at P = 90°
3. angle at P = 90°
4. it meets the line 3 cm below D
5. it meets the line 4 cm below D

★ height ≈ 4.2 cm

Examples 5 and 6 — Right-Angled Triangles and Errors

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your construction arcs

Hint construct 90°, then arc the hypotenuse

Example 5

(4 marks)

Construct a right-angled triangle with hypotenuse 6 cm and a side of 4 cm.

Example 6

(2 marks)

Ibrahim's perpendicular at P:

I compasses less than PB

II point on A: mark Q

III point on B: mark R

IV open a little wider

V bisect QR

Which two steps are wrong?

Examples 5 and 6 — Corrections

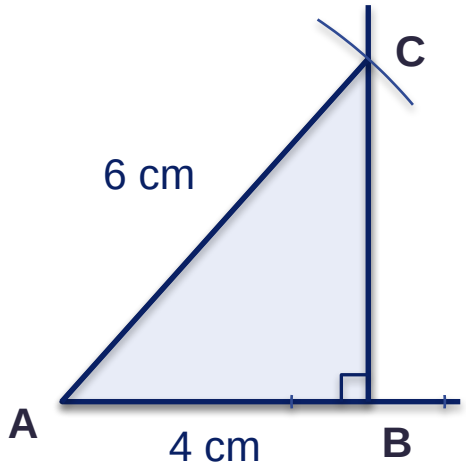
CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Construct 90° at B, then arc 6 cm from A



Tip: check: BC ≈ 4.5 cm.

Example 6

Steps II and III are wrong

The point of the compasses must go on P, not on A or B, so Q and R are the same **distance from P.**

Tip: P must be the midpoint of QR.

Task 3

PRACTICE · 9 MIN

GOLD

Questions 1–4: construct the right-angled triangle. 5–6: explain the error.

1. hypotenuse 10 cm, side 6 cm

2. hypotenuse 7 cm, side 5 cm

3. hypotenuse 9 cm, side 4 cm

4. hypotenuse 6.5 cm, side 2.5 cm

5. Maya bisects AB = 8 cm with the compasses open 3 cm. Why?

★ Leo opens the compasses wider at B than at A. What goes wrong?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **third side = 8 cm**

2. **third side ≈ 4.9 cm**

3. **third side ≈ 8.1 cm**

4. **third side = 6 cm**

5. **3 cm is less than half of 8 cm, so the arcs do not cross**

★ **the line is not the perpendicular bisector: it misses the midpoint**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Construct the perpendicular bisector of a 7 cm line.

SILVER Construct the perpendicular from a point 3 cm above a line.

GOLD Right-angled triangle: hypotenuse 5 cm, side 3 cm.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **each half 3.5 cm, angle 90°**

SILVER **it meets the line 3 cm below the point**

GOLD **third side = 4 cm**

Exam Practice — Perpendicular Bisectors

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint compasses more than half the line same width from both ends

Question 1

(2 marks)

Construct the perpendicular bisector of the line AB.



Question 2

(2 marks)

PQ is a line 9 cm long.
Construct the midpoint M of PQ.
Measure PM.

..... cm

Exam Practice — Perpendiculars

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint point on P: mark Q and R first then bisect QR

Question 3

(2 marks)

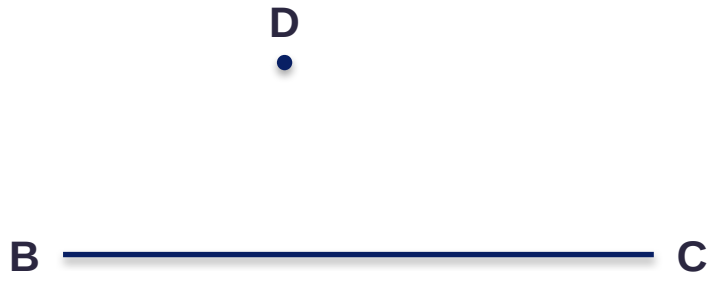
Construct the perpendicular to the line at the point P.



Question 4

(3 marks)

Construct a perpendicular from the point D to the line BC.



Exam Practice — Right Angles and Errors

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint construct 90°, then arc the hypotenuse

Question 5

(4 marks)

Construct a right-angled triangle with hypotenuse 7 cm and a side of 5 cm.

Use a ruler and compasses only.

Question 6

(2 marks)

Amir bisects AB like this:

I compasses less than half AB

II point on A: arcs above and below

III same width, point on B: arcs

IV join the crossing points

Which step is wrong? Explain.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

equal arcs from A and B — M1

line through both crossings — A1

Tip: open the compasses more than half AB.

Question 2

perpendicular bisector of PQ — M1

PM = 4.5 cm — A1

Tip: M is where the bisector crosses PQ.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

equal arcs either side of P	M1
perpendicular drawn at P	A1

Tip: the point of the compasses goes on P.

Question 4

arc from D cutting BC twice	M1
two arcs below BC	M1
line from D to BC	A1

Tip: check the angle is 90°.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

90° at one end of 5 cm	M1 M1
5 cm side measured	M1
7 cm arc to the perpendicular	A1

Tip: check: the third side ≈ 4.9 cm

Question 6

Step I	B1
more than half AB, or the arcs	
will not cross	B1

Tip: the arcs must meet above and below.